Pancancer Network Analysis reveals key Master Regulators for Cancer Invasiveness

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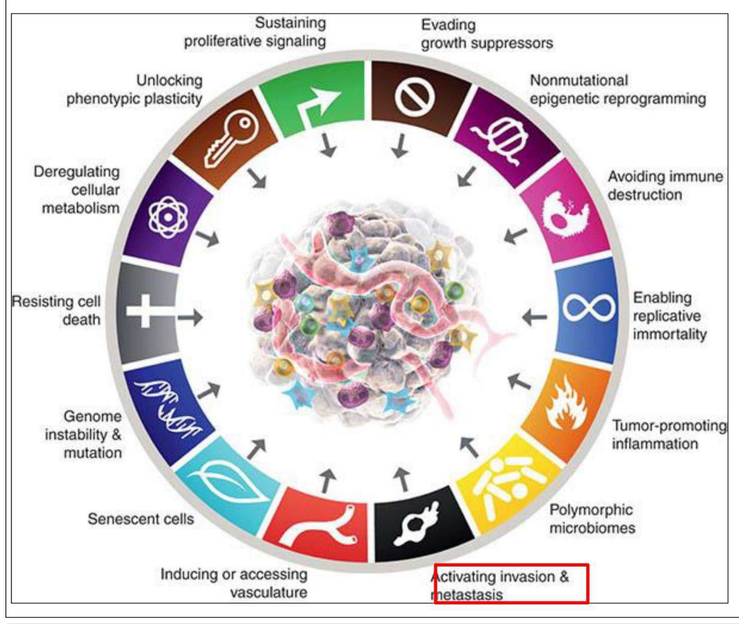
Motivations & Aims

- One hallmark of cancer is to invade local tissue or activating invasion [1].
- Tumor invasiveness reflects numerous biological changes, including tumorigenesis, progression & metastasis [2].
- Cancer invasiveness is a phenotype usually associated with worse survival prognosis [3].



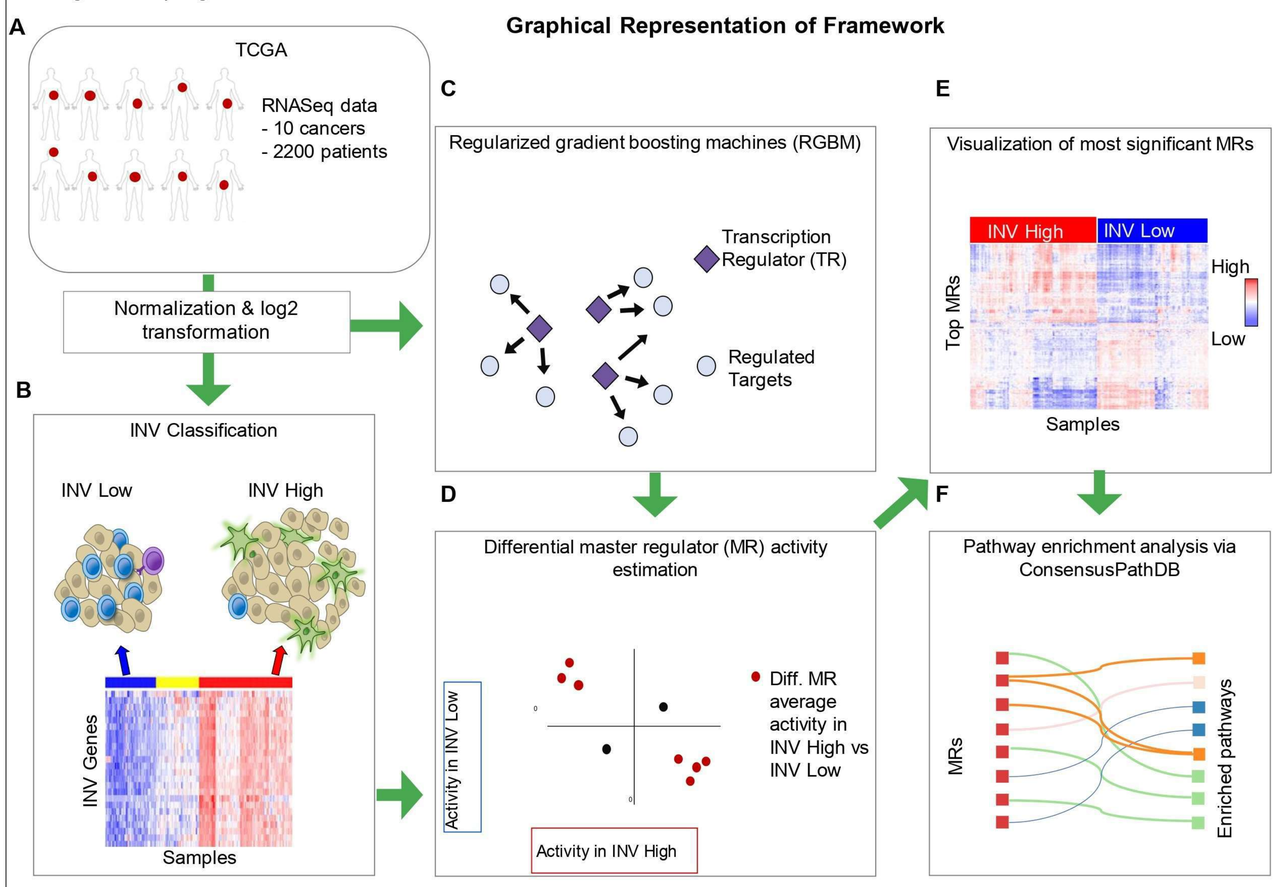
- What is the clinical impact of invasiveness for pancancer patient stratification?
- What are mechanisms governing the transcriptional regulation driving invasiveness?

- ❑ Systematic framework to determine prognostic implications of invasiveness using 24-gene set.
- ❑ Network-based framework to identify key transcriptional regulators (**master regulators**).
- ❑ Downstream analysis of master regulators (**MRs**) identify pathways sustaining oncogenic processes influencing invasiveness phenotype.
- ❑ Target **MRs & enriched pathways** for better prognosis for at risk patient populations.

Methods

- Using 24 gene-signature, we classified each sample per cancer *c* into 3 clusters i.e. Invasiveness High (INV-H), Invasiveness Medium (INV-M), and Invasiveness Low (INV-L) (Figure 1A).
- Significant prognostic association of invasiveness clusters with **overall survival for 10 out of 32** cancers in TCGA.

Graphical Representation of Framework



- **INV-H** group is always associated with **worse survival prognosis** in all the 10 cancers.
- Design a framework that takes **consensus of 4 different master regulator analysis (MRA)** pipelines to identify MRs specific to INV-H and INV-L phenotype:
  - Determine TR regulon using RNA-Seq data from TCGA [4], estimate TR regulon activities and identify statistically significant differentially activated TFs [4] referred as Master Regulators (MRs) w.r.t. INV phenotype.
  - TR-target interaction network inferred by using a non-linear ML technique called Regularized Gradient Boosting Machines (RGBM) [4,5] solving the equation Y = F(X) (Figure 1C).
  - Activity of a TR in each tumor sample can be defined as: $\mathrm{Act}^c(\mathrm{TF}, i) = \frac{1}{U}\sum_{k=1}^{U} g_{k,i}^{c,+} - \frac{1}{V}\sum_{j=1}^{V} g_{j,i}^{c,-}$
  - Differential activity of TRs estimated using limma or wilcoxon-test (Figure 1E).
  - All TRs with |NES| > 1 and FDR-adjusted p-value < 0.05 are considered as MRs for a cancer *c* (Figure 1D).
  - MRs either have high activity in INV-H samples (red) and low activity in INV-L samples (blue) or have high activity in INV-L samples and low activity in INV-H samples for a cancer *c* (Figure 1E).
- **Validation of the MR activities** specific to INV-H and INV-L on sets from two different sources, i.e. cancers where invasiveness in not prognostic and the datasets obtained from **independent PRECOG repository**.
- MRs specific to INV-H and INV-L phenotype undergo over-representation analysis using pathways derived from ConsensusPathDB [6] to identify significantly enriched pathways.

Results

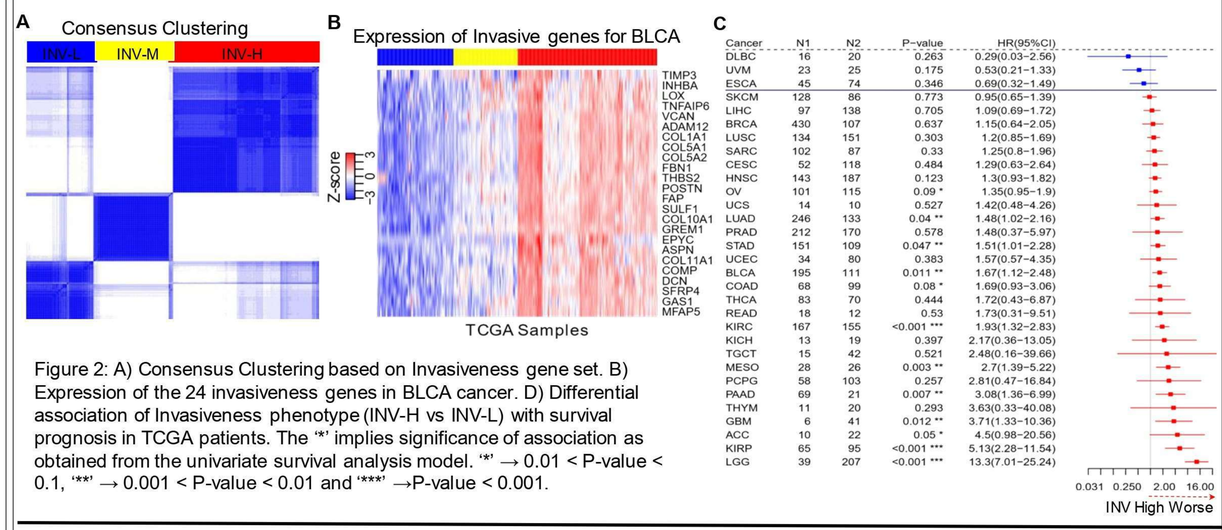


Figure 2: A) Consensus Clustering based on Invasiveness gene set. B) Expression of the 24 invasiveness genes in BLCA cancer. D) Differential association of Invasiveness phenotype (INV-H vs INV-L) with survival prognosis in TCGA patients. The '*' implies significance of association as obtained from the univariate survival analysis model. '*' → 0.01 < P-value < 0.1, '**' → 0.001 < P-value < 0.01 and '***' →P-value < 0.001.

Figure 3: MR activities and downstream pathway enrichment analysis. A) Top MRs specific to INV High/Low phenotype (MR in >5 out of 10 cancers). Significantly enriched pathways associated with MRs specific to INV High. C) Independent validation of INV-H and INV-L specific MR activity in datasets obtained from PRECOG repository. The cancer samples where invasiveness has significant association with overall survival are indicated in 'gold' whereas those with no significant association with overall survival are depicted in 'purple'.

Take Home Messages

- ➔ Invasiveness is significantly prognostic (worse survival) for 10 cancers (out of 32) in TCGA.
- ➔ Using a consensus of 4 MRA pipelines, MRs identified specific to INV-H and INV-L phenotype.
- ➔ Downstream analysis of MRs specific to INV-H lead to enriched pathways, such as the acute inflammatory response, TLR signaling, TGFβ signaling, epithelial to mesenchymal transition and extracellular matrix organization, which have connotations of **inflammatory immune response** and **feasibility for metastasis**.
- ➔ Targeting MRs and enriched pathways can possibly drive better survival for at risk patient populations.

References

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